

Operations Research 2014/2015

Date: 22/01/2015

Exam – First Call

Duration: 2 hours

Note: Present **all the calculations** that you perform and appropriately **justify** your answers.

1. Consider the following problem:

“The paper industry FAZPAPEL S.A. started its production in 2005 and it has already conquered its place in the Portuguese market, having established supply contracts for the three types of paper that it manufactures: cardboard, greaseproof paper and Kraft paper. All the production is performed in two plants, one located in Coimbra and the other in Viseu. According to the contracts, the company has to supply 8 tons of cardboard, 3 tons of greaseproof paper and 14 tons of Kraft paper. Due to the high quality of FAZPAPEL products, there is an extra demand for each type of paper that ensures the draining of any remaining production. The factory of Coimbra has a daily production cost of € 100,000.00 for a production capacity of 4 tons of cardboard, 0.5 tons of greaseproof paper and 1 ton of Kraft paper. In turn, the Viseu factory has a daily production cost of € 200,000.00 for a production capacity of 1 ton of cardboard, 0.5 tons of greaseproof paper and 3.5 tons of Kraft paper. The company wants to determine how many days each plant should operate in order to satisfy the orders at the lowest possible cost.”



To help FAZPAPEL, formulate the problem in terms of a linear programming model, indicating the meaning of the decision variables and objective function.

2. Consider the following linear programming problem:

$$\text{Minimize } z = 4x_1 + x_2$$

subject to.

$$3x_1 - x_2 \leq 6$$

$$x_1 + 3x_2 \geq 3$$

$$2x_1 + x_2 \leq 3$$

$$x_1 \geq 0, x_2 \geq 0$$

- Solve the problem by the dual Simplex method;
- Formulate the dual problem that corresponds to the one presented above;
- Without solving the dual problem, present its optimal solution as well as its objective function value;
- Comment the following sentence: “It is always preferable to solve a linear programming problem by the dual Simplex method since that way it is possible to avoid the introduction of artificial variables and the consequent employment of “Big M” or “Two Phases” techniques”.

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3. Consider now the following linear programming problem:

$$\begin{aligned} \text{Minimize } z &= -x_1 + x_2 \\ \text{subject to} \\ 5x_1 + 4x_2 &\leq 20 \\ 4x_1 + 5x_2 &\leq 20 \\ x_1 &\geq -1, x_2 \text{ free} \end{aligned}$$

- Solve it by the graphical method;
- Reformulate the problem so that all the variables have a non-negativity constraint;
- Solve the problem reformulated in b) by the Simplex method.

4. Certain food company has three factories, which we designate by **F1**, **F2**, and **F3**, where excellent quality biscuits with great acceptance in the market are made. The biscuits produced in these units follow to two distribution centres that we designate by **C1** and **C2**, which are in charge of distributing them to all the parts of the country. It is known that **F1**, **F2**, and **F3** can monthly produce **7**, **13**, and **15** tons of these biscuits, respectively, and that **C1** and **C2** require **18** and **14** tons of biscuits, per month, respectively. Follows the table with the unitary transportation costs (per ton) from each factory to each distribution centre:



		Distribution Centres	
		C1	C2
Factories	F1	7	9
	F2	6	4
	F3	5	2

The company wants to know the schema that it should use to transport the biscuits from the various factories to the distribution centres in order to minimize the total transportation cost.

- Determine an initial basic feasible solution for the problem using the Vogel's approximation method (based on penalties calculation);
- Starting from the solution obtained in a), solve the problem by the transportation method;
- Interpreting the solution achieved in b), please refer if any of the factories will have remaining biscuits or if any of the distribution centres will be left with a deficit of the same.

Scores: 1 – 3,5 points 2 – 5,5 points 3 – 5,5 points 4 – 5,5 points